

IEOR 4404: Simulation

Lecture 5: More on Acceptance- Rejection Method

A-R without Knowing Normalization Constant

Can A-R work even if we know the target density $f(x)$ only up to a normalization constant?

E.g.,

$$f(x) = \frac{1}{K} e^{-\frac{1}{2}x^2} e^{\sin^2 x}, \quad -\infty < x < \infty$$

where K is difficult to compute

This happens often in Bayesian statistics.

A-R without Knowing Normalization Constant

Suppose $f(x) = \frac{1}{K} h(x)$ where $h(x)$ is known

Suppose $g(x)$ is a proposal density, i.e., $\int_{-\infty}^{\infty} g(x) dx = 1$

A-R procedure applies analogously:

- Find a constant $\tilde{C}$ such that

$$\frac{h(x)}{g(x)} \leq \tilde{C} \text{ for all possible } x$$

- To generate one output,
 - Generate $Y \sim g(y)$
 - “Accept” Y with probability $\frac{h(Y)}{\tilde{C}g(Y)}$; otherwise “reject” and go back to the beginning

A-R without Knowing Normalization Constant

Pseudo code:

- Find a constant $\tilde{C}$ such that

$$\frac{h(x)}{g(x)} \leq \tilde{C} \text{ for all possible } x$$

- To generate one output,
 - Generate $Y \sim g(y)$
 - Generate $U \sim Unif(0,1)$
 - If $U \leq \frac{h(Y)}{\tilde{C}g(Y)}$, set $X = Y$ (accept); otherwise, go back to the beginning (reject)
 - Output X

Example

- Generate a r.v. with pdf

$$f(x) = \frac{1}{K} e^{-\frac{1}{2}x^2} e^{\sin^2 x}, \quad -\infty < x < \infty$$

with an unknown K

- Use standard normal as the proposal

$$g(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty$$

$$f(x) = \frac{1}{k} e^{-\frac{x^2}{2}} e^{\sin^2 x} \quad \text{for } -\infty < x < \infty$$

$$g(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for } -\infty < x < \infty$$

Find $\tilde{C}$ such that

$$\frac{h(x)}{g(x)} \leq \tilde{C} \quad \text{for } -\infty < x < \infty$$

$$\frac{h(x)}{g(x)} = \frac{e^{-\frac{x^2}{2}} e^{\sin^2 x}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}}$$

$$= \sqrt{2\pi} e^{\sin^2 x}$$

$$\leq \sqrt{2\pi} e$$

We can choose $\tilde{C} = \sqrt{2\pi} e$

A - C:

① Generate $Y \sim N(0, 1)$

② "Accept" Y w.p. $\frac{h(Y)}{\tilde{c}g(Y)}$

$$= \frac{e^{-\frac{Y^2}{2}} e^{\sin^2 Y}}{\sqrt{2\pi} e \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{Y^2}{2}}} = e^{\sin^2 Y - 1}$$

otherwise reject and go back to ①

pseudo-code:

- ① Generate $Y \sim N(0, 1)$
- ② Generate $U \sim \text{Unif}(0, 1)$
- ③ If $U \leq \frac{h(Y)}{\tilde{c}g(Y)} = e^{\beta \gamma^2 Y - 1}$, set $X = Y$
otherwise go back to ①

Output X .

$$f(x) = \frac{1}{K} h(x)$$

target pdf

$$g(x)$$

proposal pdf

"standard" A-R:

Find constant C such that

$$\frac{f(x)}{g(x)} \leq C$$

$$\Rightarrow \frac{h(x)}{K g(x)} \leq C$$

Note that $\tilde{C}$ is a constant such that

$$\frac{h(x)}{g(x)} \leq \tilde{C}$$

$$\Rightarrow \tilde{C} = CK \quad \text{or} \quad C = \frac{\tilde{C}}{K}$$

In "standard" A-R:

Accept γ w.p.

$$\frac{f(\gamma)}{C_g(\gamma)} = \frac{h(\gamma)}{K C_g(\gamma)}$$
$$= \frac{h(\gamma)}{\tilde{C}_g(\gamma)}$$

which matches the procedure in the new A-R.

Why Does It Work

In the described procedure,

- Our constant $\tilde{C}$ satisfies

$$\frac{f(x)}{g(x)} = \frac{h(x)}{Kg(x)} \leq \frac{\tilde{C}}{K} \text{ for all possible } x$$

That is, the C in the standard A-R is $\frac{\tilde{C}}{K}$

- Our acceptance probability is $\frac{h(Y)}{\tilde{C}g(Y)} = \frac{h(Y)}{CKg(Y)} = \frac{f(Y)}{cg(Y)}$

Why Does It Work

- The output matches the standard A-R procedure, without us needing to know K
- The K is “canceled out” in $f(x)$ and C in the acceptance probability

Estimating Normalization Constant

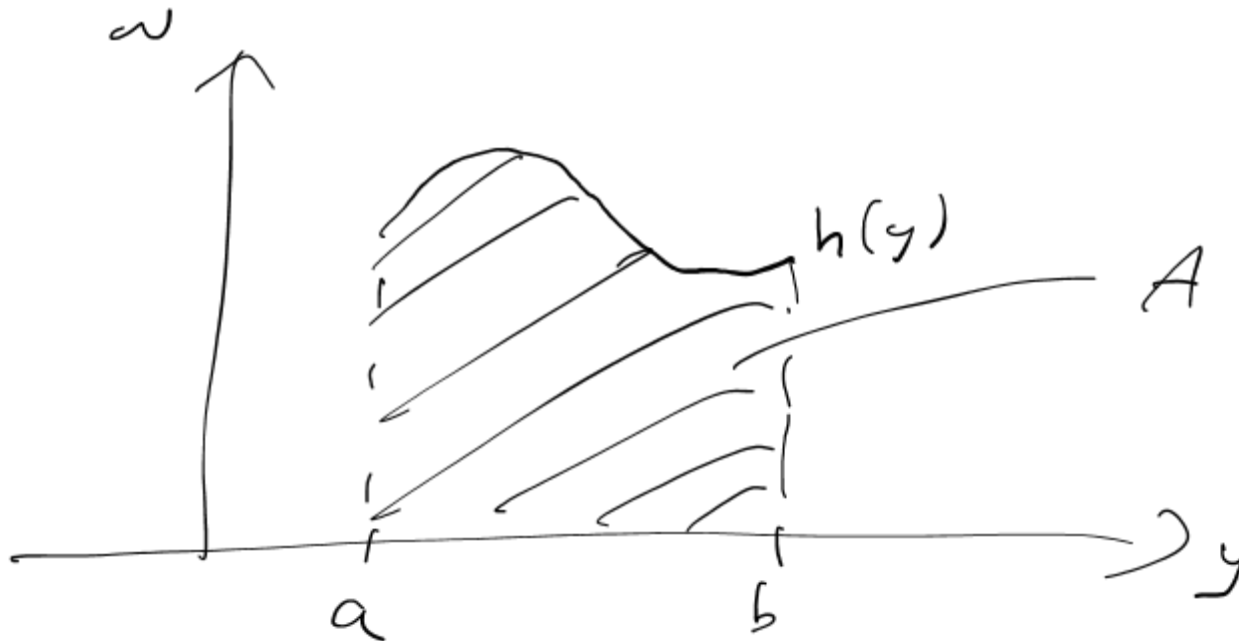
- The constant K is sometimes important
- Note that $C = \frac{\tilde{C}}{K}$, so $K = \frac{\tilde{C}}{C}$
- $C = \frac{1}{p} = \text{expected runtime}$, so $K = \frac{\tilde{C}}{\text{expected runtime}} = \tilde{C}p$
- To estimate K , record the frequency of acceptance $\hat{p}$ and use $\tilde{C}\hat{p}$

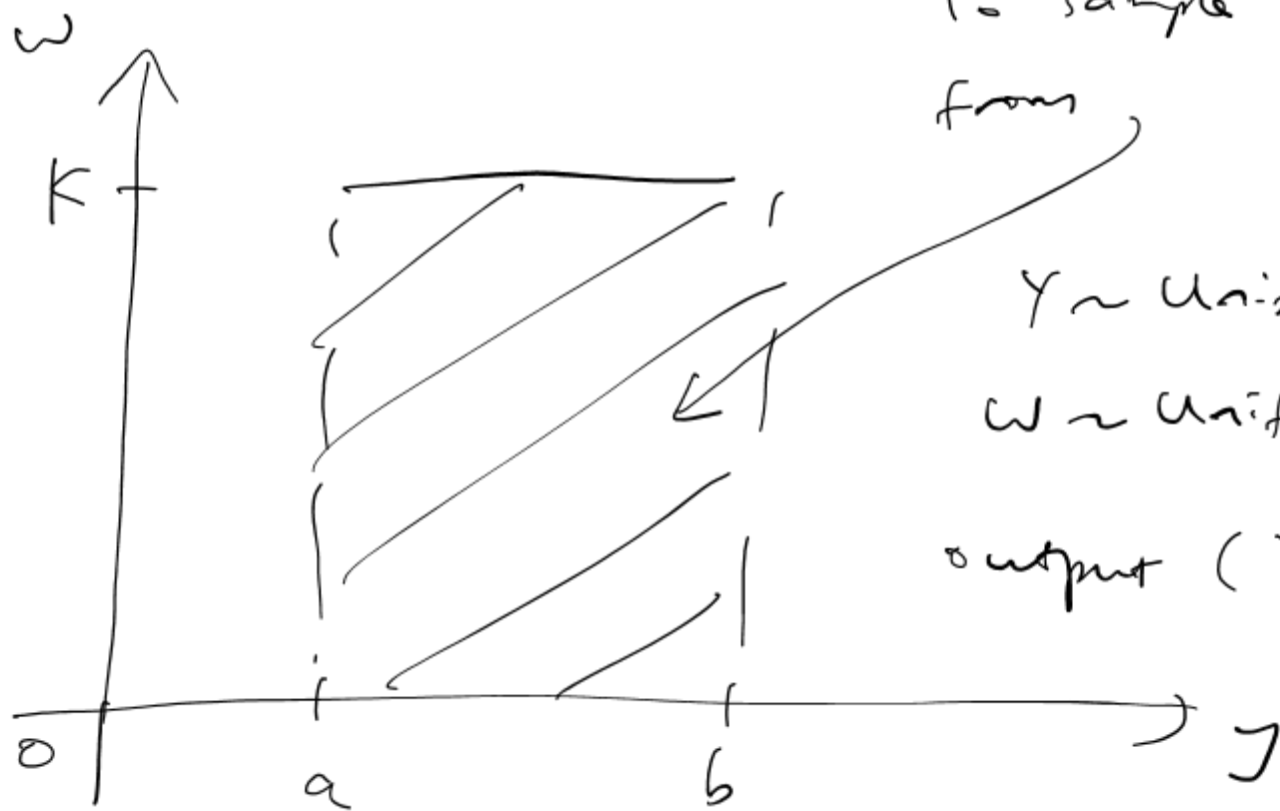
Uniform Sampling on Irregular Regions

We want to uniformly sample from region

$$A = \{(y, w): a \leq y \leq b, 0 \leq w \leq h(y)\}$$

for some function $h(y)$



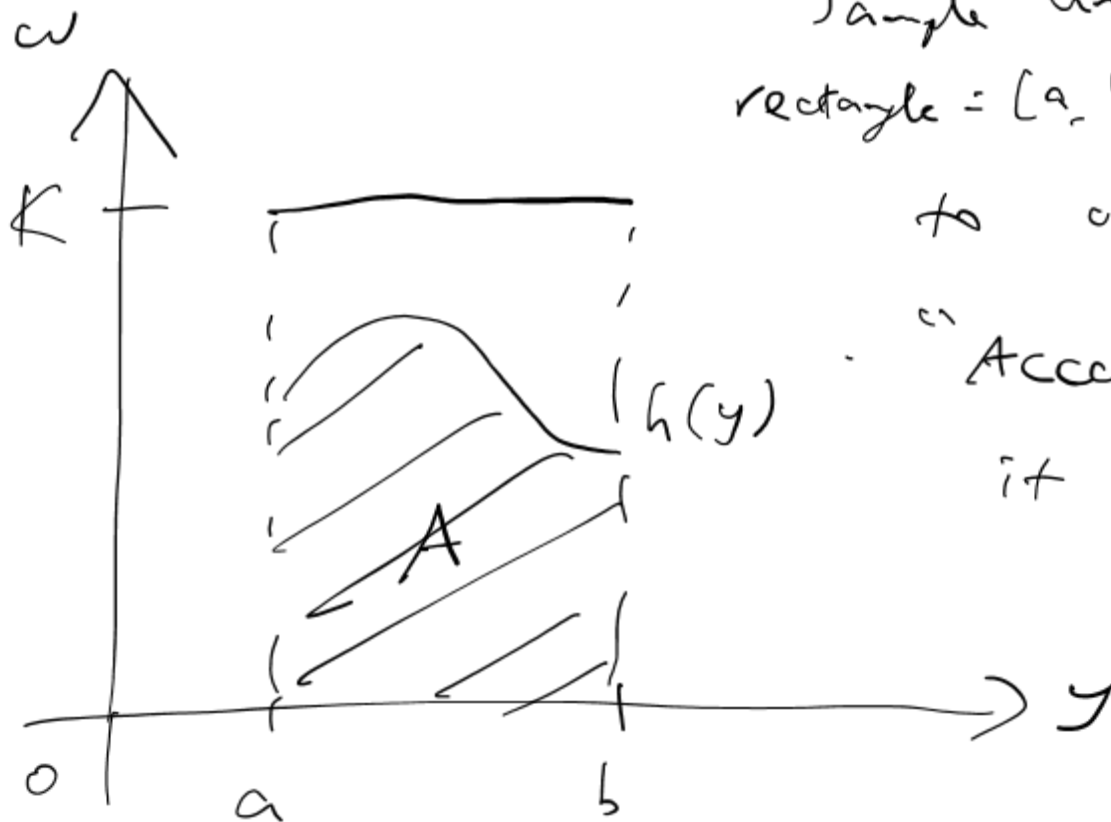


To sample uniformly
from

$$Y \sim \text{Unif}(a, b)$$

$$w \sim \text{Unif}(0, K)$$

output (Y, w)



• Sample uniformly on
rectangle $= [a, b] \times [0, K]$

to obtain (Y, ω)

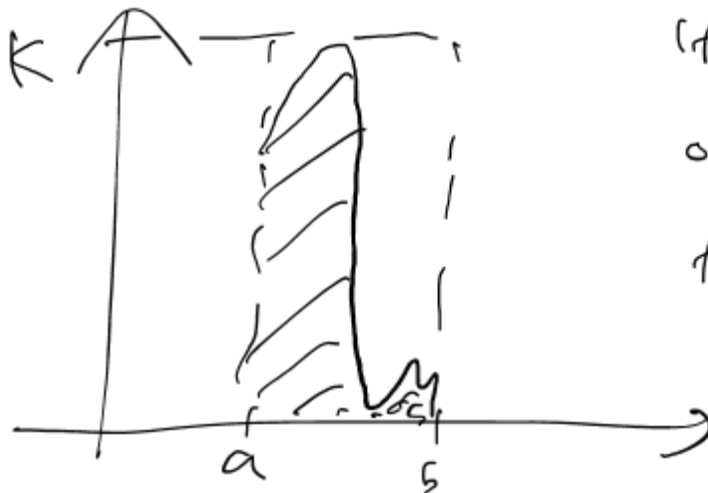
• "Accept" (Y, ω) if
it falls into A .

Uniform Sampling on Irregular Regions

If $h(y) \leq K$, then an approach:

- Sample (Y, W) uniformly in $[a, b] \times [0, K]$
- If $(Y, W) \in A$, “accept” (Y, W) ; otherwise, “reject” and go back to the above

This approach could be slow. Why?



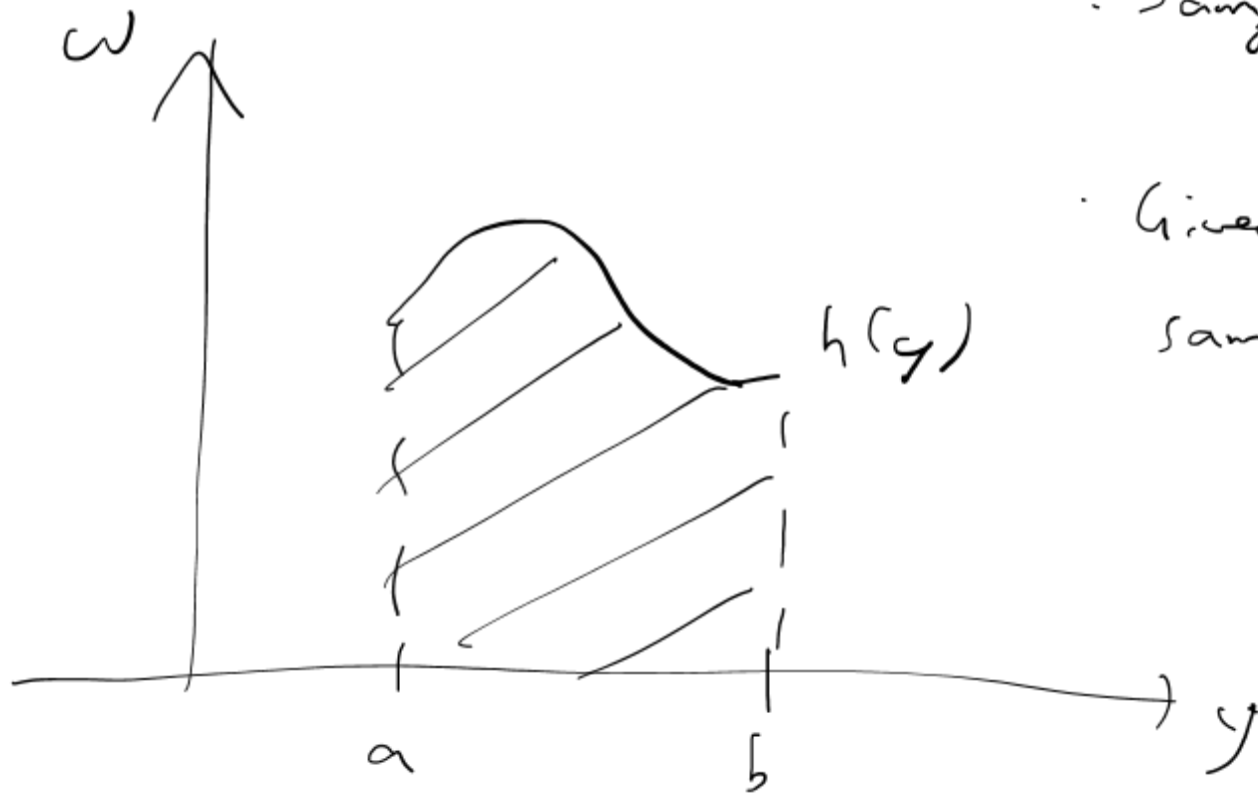
if the rectangle has a lot
of “rejection” area,
then $P(\text{acceptance}) = \text{low}$
 $\Rightarrow$ algo is slow.

Uniform Sampling on Irregular Regions

Alternative approach:

- Sample Y from $\frac{h(y)}{K}$, $a \leq y \leq b$, where K is a (possibly unknown) normalization constant
- Given Y , sample W from $Unif(0, h(Y))$

This approach is faster, assuming we can generate Y from $\frac{h(y)}{K}$ efficiently



• Sample $Y \sim \frac{h(y)}{K}$

• Given Y ,
sample $w \sim \text{Unif}(0, h(y))$

Uniform Sampling on Irregular Regions

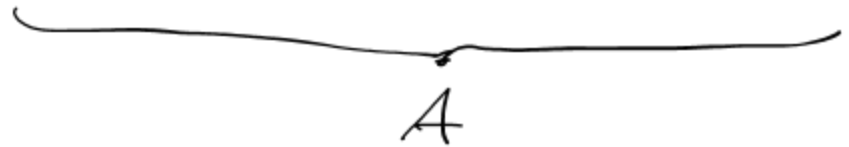
Why does the alternative approach work?

- Joint pdf:

$$f_{Y,W}(y,w) = \frac{1}{K}, \quad a \leq y \leq b, 0 \leq w \leq h(y)$$

← uniform density on A .

where K is the area of the region $\{(y,w): a \leq y \leq b, 0 \leq w \leq h(y)\}$



- Marginal pdf of Y :

$$f_Y(y) = \int_0^{h(y)} \frac{1}{K} dw = \frac{h(y)}{K}, \quad a \leq y \leq b$$

- Conditional pdf of W given Y :

$$f_{W|Y}(w|y) = \frac{f_{Y,W}(y,w)}{f_Y(y)} = \frac{1/K}{h(y)/K} = \frac{1}{h(y)}, \quad a \leq y \leq b, 0 \leq w \leq h(y)$$

Uniform Sampling on Irregular Regions

Requirement: We need K to be finite, i.e.,

$$K = \int_a^b \int_0^{h(y)} dw dy = \int_a^b h(y) dy < \infty$$

Relation with A-R

Suppose $h(y) \leq K$

Then, the naïve approach:

- Generate $Y \sim \text{Unif}(a, b)$
- Generate $W \sim \text{Unif}(0, K)$
- Accept if $W \leq h(Y)$

"standard" A-R:

← generate from proposal

← coin toss randomness

← accept-reject procedure

Relation with A-R

Suppose $h(y) \leq K$

Then, the naïve approach:

- Generate $Y \sim \text{Unif}(a, b)$
- Generate $U \sim \text{Unif}(0, 1) \leftarrow w = uK$
- Accept if $U \leq \frac{h(Y)}{K} \leftarrow w \leq h(Y)$

$\left. \begin{array}{l} \text{- target pdf} = \frac{h(y)}{K} \text{ or } \frac{h(y)}{\text{constant}} \\ \text{- proposal pdf} = \text{unif}(a, b) \end{array} \right\} \begin{array}{l} \Rightarrow u \leq \frac{h(Y)}{K} \\ \text{This A-R is exactly} \\ \text{as above.} \end{array}$

Using the new A-R (without knowing normalization constant):

We find constant $\tilde{C}$ such that

$$\frac{h(y)}{g(y)} \leq \tilde{C}$$

$$\Rightarrow \frac{h(y)}{\left(\frac{1}{b-a}\right)} \leq \tilde{C}$$

$$\Rightarrow h(y) \leq \tilde{C}(b-a) = K$$

Relation with A-R

Equivalent to an A-R procedure that uses proposal density $Unif(a, b)$ to generate a target density $h(y)$ up to a normalization constant

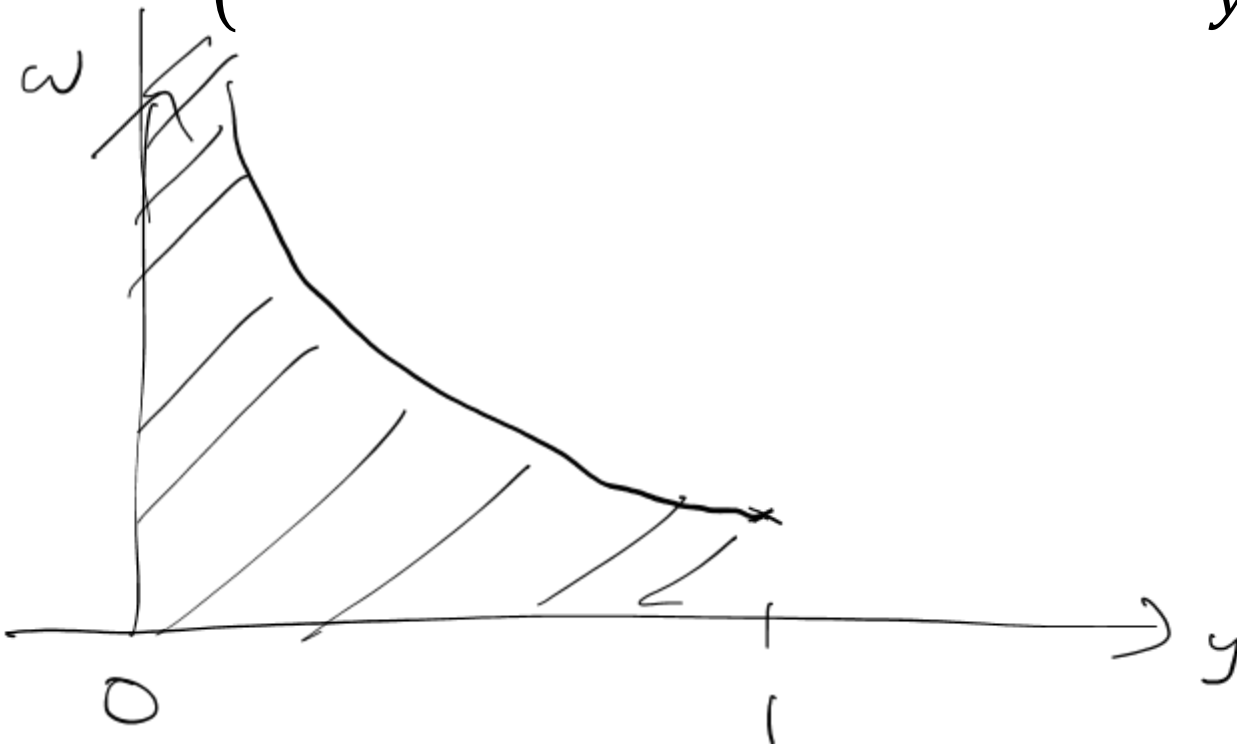
By this equivalence, we know Y has marginal density $h(y)$ up to a normalization constant

The faster approach we describe is to generate this Y directly

Example

Sample (Y, W) uniformly in the region

$$A = \left\{ (y, w) : 0 \leq y \leq 1, 0 \leq w \leq \frac{\cos y + 1}{y^{\frac{1}{4}}} \right\}$$



1st approach (A-R) fails because we cannot contain A inside any rectangle.

We need to use 2nd approach:

$$\text{area of } A = K = \int_0^1 h(y) dy < \infty$$

$$= \int_0^1 \frac{\cos y + 1}{y^{\frac{1}{4}}} dy \leq \int_0^1 \frac{2}{y^{\frac{1}{4}}} dy = \frac{8}{3} y^{\frac{3}{4}} \Big|_0^1 = \frac{8}{3}$$

• Sample $\gamma \sim \frac{h(\gamma)}{k} = \frac{\cos \gamma + 1}{k \gamma^{\frac{1}{4}}}$

• Given γ , sample $W \sim \text{Unif}(0, h(\gamma))$

$$\text{Unif}\left(0, \frac{\cos \gamma + 1}{\gamma^{\frac{1}{4}}}\right)$$

To execute the first sampling step, i.e.

$$\text{sample } Y \sim \frac{\cos y + 1}{4 y^{\frac{1}{4}}}, \quad 0 < y < 1.$$

use A-R (new version)

Consider proposal density

$$g(y) = \frac{3}{4 y^{\frac{1}{4}}}, \quad 0 < y < 1$$

$$\text{Since } \int_0^1 \frac{1}{y^{\frac{1}{4}}} dy = \left. \frac{4}{3} y^{\frac{3}{4}} \right|_0^1 = \frac{4}{3}$$

A-R (new version):

Find $\tilde{C}$ such that

$$\frac{h(y)}{g(y)} \leq \tilde{C} \quad \text{for } 0 < y < 1$$

$$\Rightarrow \frac{\frac{\cos y + 1}{y^{\frac{1}{4}}}}{\frac{3}{4y^{\frac{1}{4}}}} = \frac{4}{3}(\cos y + 1) \leq \frac{8}{3}$$

$\Rightarrow$ choose $\tilde{C} = \frac{8}{3}$

So the proposal $g(y)$ is valid.

① Generate $Y \sim g(y) = \frac{3}{4} y^{-\frac{1}{4}}$

$$\begin{aligned} \text{(by ITM: cdf)} &= \int_0^y \frac{3}{4} z^{-\frac{1}{4}} dz = z^{\frac{3}{4}} \Big|_0^y \\ &= y^{\frac{3}{4}} \end{aligned}$$

$$\text{Set } u = y^{\frac{3}{4}} \Rightarrow y = u^{\frac{4}{3}}$$

1 TM: Generate $u \sim \text{Unif}(0, 1)$
 output $u^{\frac{4}{3}}$

② Generate $V \sim \text{Unif}(0, 1)$

③ if $V \leq \frac{h(Y)}{\tilde{C}g(Y)} = \frac{\frac{4}{3}(\cos Y + 1)}{\frac{4}{3}}$

$= \frac{1}{2}(\cos Y + 1)$, set $X = Y$
 otherwise go back to ①.